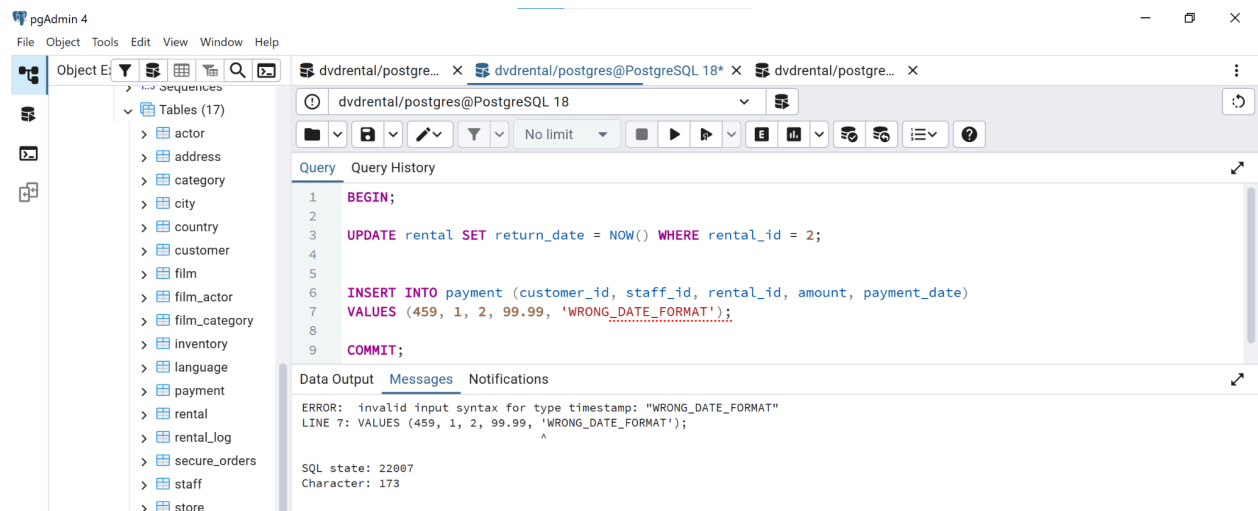
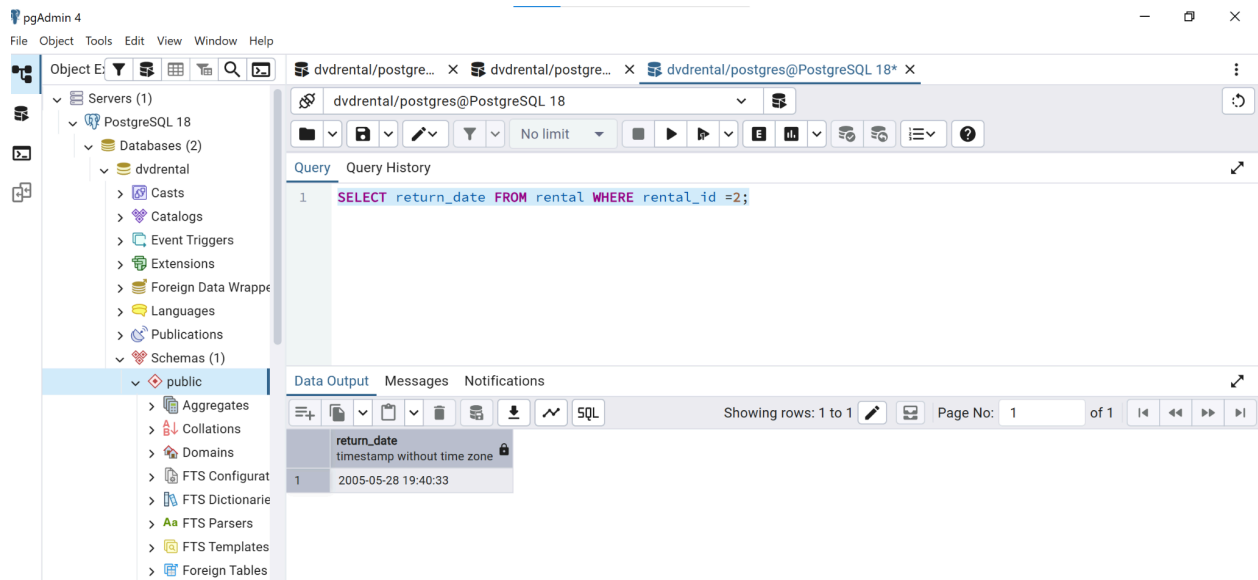


Name: Bùi Quang Minh

CLASS: DS66b

Ex1:



No, the return_date is not updated. Because the second statement fails (due to invalid date format), PostgreSQL aborts the entire transaction block to maintain Atomicity ("All or Nothing").

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store
- Trigger Functions
- Types

Query Editor

Query: dvddental/postgres@PostgreSQL 18

```
1 BEGIN;
2 INSERT INTO customer (first_name, last_name, email, address_id, store_id)
3 VALUES ('AI', 'Student', 'ai@edu.vn', 1, 1);
4
5 SAVEPOINT user_created;
6
7 INSERT INTO special_offers (customer_id, code) VALUES (999, 'FREE');
8 ROLLBACK TO user_created;
9
10 COMMIT;
11
```

Data Output Messages Notifications

SAVEPOINT

Query returned successfully in 92 msec.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store
- Trigger Functions
- Types

Query Editor

Query: dvddental/postgres@PostgreSQL 18

```
1 BEGIN;
2 INSERT INTO customer (first_name, last_name, email, address_id, store_id)
3 VALUES ('AI', 'Student', 'ai@edu.vn', 1, 1);
4
5 SAVEPOINT user_created;
6
7 INSERT INTO special_offers (customer_id, code) VALUES (999, 'FREE');
8 ROLLBACK TO user_created;
9
10 COMMIT;
11
```

Data Output Messages Notifications

ERROR: relation "special_offers" does not exist
LINE 1: INSERT INTO special_offers (customer_id, code) VALUES (999, ...
^

SQL state: 42P01
Character: 13

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store

Query Editor: dvdrental/postgres@PostgreSQL 18

Query

```
1 BEGIN;
2 INSERT INTO customer (first_name, last_name, email, address_id, store_id)
3 VALUES ('AI', 'Student', 'ai@edu.vn', 1, 1);
4
5 SAVEPOINT user_created;
6
7 INSERT INTO special_offers (customer_id, code) VALUES (999, 'FREE!');
8 ROLLBACK TO user_created;
9
10 COMMIT;
11
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 114 msec.

Check again:

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store
- Trigger Functions

Query Editor: dvdrental/postgres@PostgreSQL 18

Query

```
1 SELECT * FROM customer WHERE email = 'ai@edu.vn';
```

Data Output Messages Notifications

Showing rows: 1 to 1 of 1

	customer_id [PK] integer	store_id smallint	first_name character varying (45)	last_name character varying (45)	email character varying (50)	address_id smallint	activebool boolean	create_date date	last_update timestamp
1	601	1	AI	Student	ai@edu.vn	1	true	2026-09-18	2026-09-18

EX2:

Before:

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store
- Trigger Functions
- Types

Query

```
select * from staff;
```

Query History

Data Output

Showing rows: 1 to 2 of 1

	staff_id [PK] integer	first_name character varying (45)	last_name character varying (45)	address_id smallint	email character varying (50)	store_id smallint	active boolean	username character varying (16)
1	1	Mike	Hillyer	3	Mike.Hillyer@sakilastaff.com	1	true	Mike
2	2	Jon	Stephens	4	Jon.Stephens@sakilastaff.c...	2	true	Jon

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store

Query

```
BEGIN;  
  
INSERT INTO staff (first_name, last_name, address_id, store_id, username)  
VALUES ('Binh', 'Tran', 1, 2, 'binh_tran');  
SAVEPOINT emp_created;  
  
UPDATE staff SET store_id = 'INVALID' WHERE username = 'binh_tran';
```

Query History

Data Output

Messages

Notifications

SAVEPOINT

Query returned successfully in 115 msec.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store
- Trigger Functions
- Types
- Views

Query Editor: dvdrental/postgres@PostgreSQL 18*

Query

```
1 BEGIN;
2
3 INSERT INTO staff (first_name, last_name, address_id, store_id, username)
4 VALUES ('Binh', 'Tran', 1, 2, 'binh_tran');
5 SAVEPOINT emp_created;
6
7 UPDATE staff SET store_id = 'INVALID' WHERE username = 'binh_tran';
```

Data Output Messages Notifications

ERROR: invalid input syntax for type smallint: "INVALID"
LINE 1: UPDATE staff SET store_id = 'INVALID' WHERE username = 'binh...

SQL state: 22P02
Character: 29

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store
- Trigger Functions
- Types
- Views

Query Editor: dvdrental/postgres@PostgreSQL 18*

Query

```
1 BEGIN;
2
3 INSERT INTO staff (first_name, last_name, address_id, store_id, username)
4 VALUES ('Binh', 'Tran', 1, 2, 'binh_tran');
5 SAVEPOINT emp_created;
6
7 UPDATE staff SET store_id = 'INVALID' WHERE username = 'binh_tran';
8
9 ROLLBACK TO emp_created;
10 UPDATE staff SET store_id = 1 WHERE username = 'binh_tran';
11 COMMIT;
```

Data Output Messages Notifications

COMMIT

Query returned successfully in 96 msec.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Sequences
- Tables (17)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - film
 - film_actor
 - film_category
 - inventory
 - language
 - payment
 - rental
 - rental_log
 - secure_orders
 - staff
 - store
- Trigger Functions
- Types
- Views

Query Editor: dvdrental/postgres@PostgreSQL 18*

Query

```
1 SELECT first_name, last_name, store_id, username
2 FROM staff
3 WHERE username = 'binh_tran';
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	first_name character varying (45)	last_name character varying (45)	store_id smallint	username character varying (16)
1	Binh	Tran	1	binh_tran

